ENGINEERING PROJECT I

On

Robust Gait Pattern Classification Using a Random Forest Classifier and Bilateral Plantar Pressure Features

A Project Report Submitted to Central University of Jharkhand for the Partial fulfilments of the requirement for the degree of

INTEGRATED B. TECH AND M. TECH

IN

COMPUTER SCIENCE AND ENGINEERING

WITH SPECIALIZATION IN MACHINE LEARNING AND DATA SCIENCE

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CERTIFICATE

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This is to certify that the project report entitled Robust Gait Pattern Classification Using a Random Forest Classifier and Bilateral Plantar Pressure Features submitted by Md. Tarik Anvar and Nishan Das in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of work carried out by them under my supervision and guidance.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree or diploma.

Dr. Prashant Prashun Associate Professor (Project Guide)

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ACKNOWLEDGEMENT

We wish to express our profound gratitude and sincere thanks to our project guide, Dr. Prashant Prashun, Associate Professor, Department of Computer Science and Engineering, Central University of Jharkhand, for his invaluable guidance, constant encouragement, and insightful suggestions throughout this project work. His mentorship was instrumental in the successful completion of this report.

We would also like to thank the Head of the Department and all the faculty members of the Department of Computer Science and Engineering for providing us with a conducive environment and the necessary facilities to carry out this work.

Finally, we are deeply indebted to our parents and family members for their unfailing love, support, and encouragement, which has been our pillar of strength.

Nishan Das
Md. Tarik Anvar

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ABSTRACT

Human gait serves as a crucial biomarker for neuromuscular and cognitive health, with alterations in walking patterns often signaling the early onset of conditions such as neuropsychiatric disorders. Since clinical gait analysis traditionally relies on laboratory-based systems like optical motion capture and force plates, which are expensive and impractical for continuous monitoring, this work motivates the development of cost-effective, wearable solutions[1] capable of delivering clinically relevant gait metrics in real-world settings. To address this, we propose a low-cost, Force Sensitive Resistor (FSR)-based wearable system designed for the robust classification of normal and abnormal gait patterns. Each custom-designed insole is equipped with four FSRs—three positioned in the forefoot region to capture metatarsal pressure distribution and one under the heel for detecting heel-strike events—where sensor outputs are processed through voltage divider circuits and digitized via an Arduino Uno before transmission for analysis. A bilateral feature engineering pipeline is utilized to extract key gait metrics, including cadence, gait asymmetry index, and the coefficient of variation of step time, from synchronized left- and right-foot data to train a Random Forest classifier. We hypothesize that the integration of strategically placed FSRs with bilateral feature engineering will enable the detection of subtle gait anomalies while minimizing hardware complexity. This study aims to demonstrate that an affordable, wearable plantar pressure sensing system, when combined with advanced feature engineering and machine learning, can achieve clinically meaningful performance, potentially paving the way for scalable, accessible, and continuous gait monitoring solutions in healthcare, rehabilitation tracking, and early screening for movement disorders.

1.INTRODUCTION

1.1Human Gait

Human gait refers to how a person moves their limbs during locomotion, whether naturally or through specialized training. It involves bipedal forward propulsion of the center of gravity with alternate sinuous movements of different body segments, while expending the least amount of energy. Gait kinematics can differ depending on an individual's emotional state. In clinical settings, observing gait patterns can help detect movement abnormalities and provide diagnostic clues for neurological disorders such as cerebral palsy, Parkinson's disease, and Rett syndrome at an early stage.

1.2Pronation and Supination

The terms pronation and supination are used to describe how a person stands, walks, and runs. It is important to balance weight on both feet while moving, without leaning the foot in or out. Pronation and supination are terms which can be defined as the natural movements that help absorb impact while taking steps. However, excessive supination can cause stress on the outer side of the foot, resulting in uneven wear on the shoes outside sole and strain on the ankles. This leads to shin splints, calluses, bunions, and discomfort in the heels and balls of the feet. This is also known as under-pronation. On the other hand, excessive pronation, commonly referred as overpronation, occurs when the foot rolls inwards causing the arch to flatten out, and can result in uneven wear on the inside sole of the shoe. Overpronation is a more prevelant condition than under- pronation.

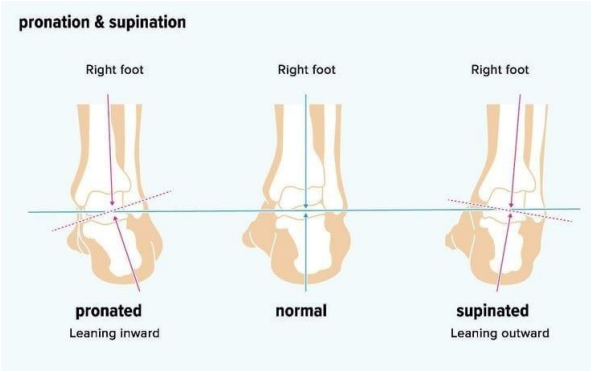

Fig 1.1 Pronation and Supination in Right Foot

In Xu Shihao et al.'s study[6], they define human gait as the specific way individuals walk. It is considered a significant and routine motion that provides observers with valuable insights into the walker's characteristics and behaviors.

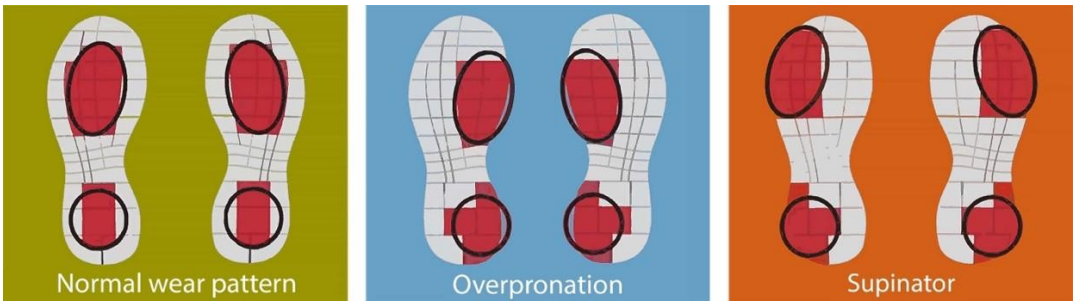

Fig 1.2 Weight Pressure region on the sole of the shoe

1.3 Gait and its Relation to Human Behavior

Gait is an example of behaviour that reflects various levels of nervous system function. In psychiatric conditions, gait disturbances are thought to reflect defective brain function. Patients who suffer from gait disturbances tend to develop balance disorders as well as impaired body posture. They have included three major mental illnesses Depression, Anxiety and Schizophrenia. They found that the patients suffering from mental disorders reveal a distinctive physical profile that is in accordance with the clinical diagnosis. The physical profile of patients shows the following symptoms: for schizophrenia, the patients exhibited slow gait and decreased stride length, while the patients suffering from anxiety disorders showed balance disorders, and those suffering from depression exhibited slow gait followed by a slumped posture[2]. In our research, we are mainly focusing on the balance disorders that happen due to anxiety disorders

1.4 Different Types of Gait Data

1.Multi-camera, Marker-based systems:

Cutting-edge technology for gait analysis is typically found in specialized laboratories and large clinical centres. These systems involve the use of reflective markers, or sensors, which are worn by the subject to capture joint movements with great accuracy. Force plates and electromyography (EMG) are also commonly used alongside these systems to record muscle activity. However, these tools are limited to lab use.

2.Pressure-Insoles, EMG, EEG:

Force plates are utilized to measure Ground Reaction Forces (GRFs) by placing sensors on the floor or treadmill to assess the pressure generated during a subject's walking motion. Many gait analysis studies consider the forces applied to the surface because they correspond to the load placed on the joints. To measure GRFs, wearable pressure insoles can be used as an alternative to force plates. Goniometers, which use flex sensors, are another tool employed to measure joint angles, including those of the ankles, knees, and hips.

3.Markerless, RGB & RGBD systems:

Conversely, vision-based systems that are markerless and utilize RGB cameras and RGB-D cameras do not necessitate the us of specialized equipment. Advanced computer vision algorithms have been created to extract crucial joints in real-time and predict gait attributes. These systems can effectively be employed for gait analysis, activity recognition, and energy expenditure estimation. However, they pose privacy concerns for subjects, which has limited their acceptance.

4.Single IMU gait sensing:

Recently, Inertial Measurement Units (IMUs) sensors have proven to be a helpful tool for studying human gait. These sensors can be affixed to any part of the body and utilize a combination of accelerometers, gyroscopes, and magnetometers to measure velocity, acceleration, and orientation in relation to gravitational forces. To promote patient compliance and simplify use in home environments, single sensors have been developed to capture gait data and identify walking gait impairments.

1.5Applications of Gait Analysis

When we go in-depth about what the gait assessment of a human can provide us, there are various applications which we can think of. Through gait assessment, we can understand the emotional as well as the physical abnormalities.

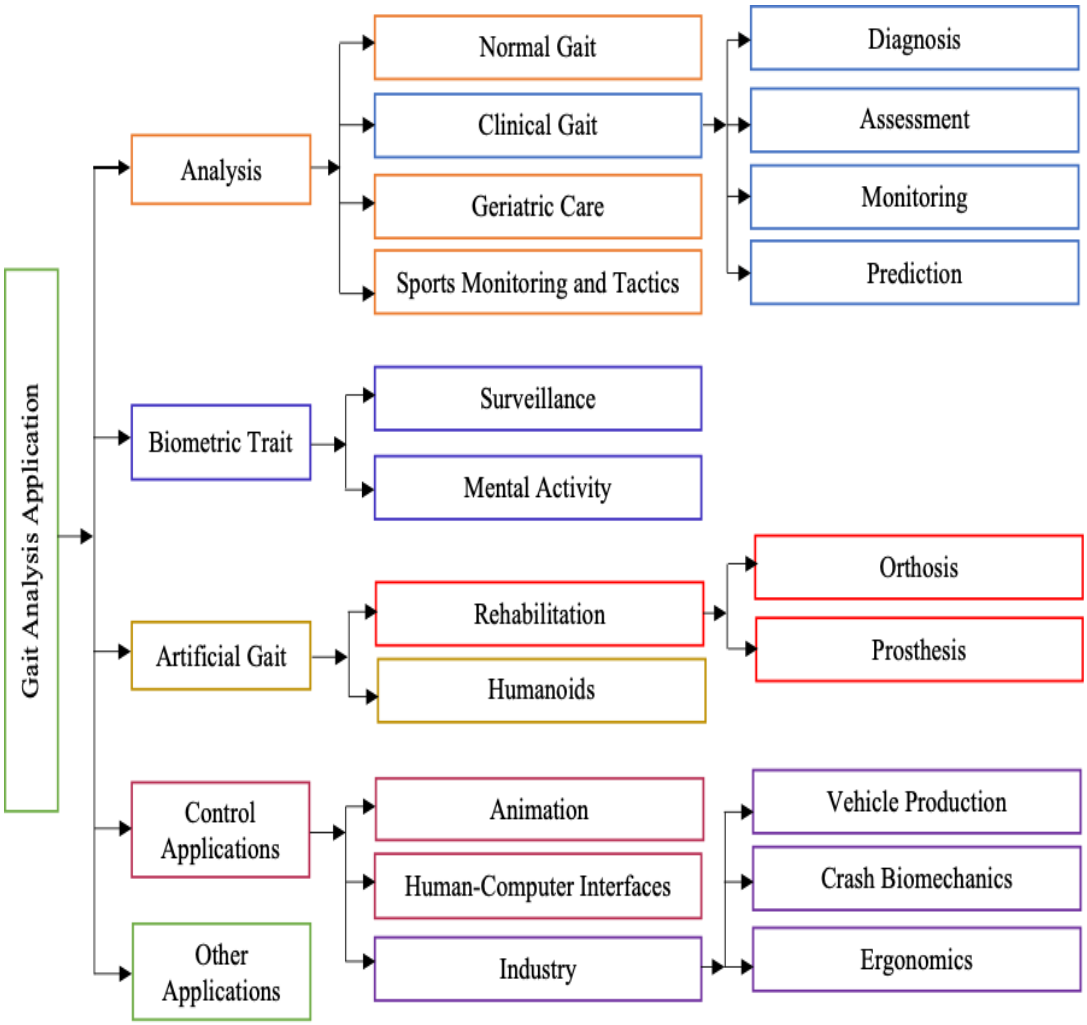

Fig 1.3: Gait Analysis Application

1.6 Problem Statement and Objective

- Human gait entails significant information about an individual's emotional state.
- Researchers agree that anxiety is related to balance dysfunction.
- They also demonstrated that treatment targeting balance problems mitigates dysfunction, greatly reduces anxiety and increases self-esteem.
- Various gait data are used to predict an individual's behavioral aspects.
- One such gait data is through the kinematics of the lower limb.
- Our goal is to determine if we can use "foot pressure data" to predict the emotional state of a human being using Random Forest.
- By doing so, we can help the individual to wear the right kind of shoes to elicit balance in their behavioral states.

2. LITERATURE REVIEW

2.1 Studies indicating the relationship between human behavioral aspects and their gait

Extensive research has investigated the correlation between biomechanical gait patterns and psychological well-being, specifically focusing on the detection of anxiety and depressive disorders. The primary objective of these studies is to determine whether human kinematics can serve as a biomarker for internal emotional states.

To facilitate this analysis, researchers have employed depth-sensing imaging technology to record locomotion. This approach offers a significant advantage as a non-intrusive, contactless method, eliminating the need for wearable sensors or wired apparatuses, thereby allowing participants to exhibit natural, uninhibited walking patterns.

To validate the efficacy of this methodology, the gait data collected was cross-referenced with standardized mental health assessments. Validation protocols included the administration of validated psychometric questionnaires and clinical diagnoses provided by psychiatrists. By processing this kinematic data through advanced computational algorithms, distinct behavioral patterns emerged. The results demonstrated that machine learning models could effectively identify individuals suffering from depression or anxiety based solely on posture and stride analysis. Collectively, these findings substantiate the existence of a tangible relationship between human locomotion and psychological state, suggesting that gait analysis offers a viable, non-invasive modality for the diagnosis and monitoring of mental health disorders.

2.2 Relationship between Gait and Balance of the Human Body

Scientific inquiry has also examined the hypothesis that improvements in physical stability can positively influence psychological health. Notably, research focused on pediatric populations has investigated whether correcting balance deficits acts as a mechanism to mitigate anxiety and enhance self-esteem.

Initial baseline assessments revealed a significant correlation between motor proficiency and emotional state: children exhibiting deficits in physical coordination demonstrated markedly higher levels of anxiety and lower self-esteem compared to a control group with normal balance function. To address this, a longitudinal balance training regimen was implemented. Post-intervention analysis indicated that as participants improved their postural stability, there was a concurrent and

significant improvement in their psychological profile, characterized by reduced anxiety and elevated self-worth.

These findings were further corroborated by clinical observational studies conducted in hospital settings. Through the administration of standardized physical assessments—such as the Unipedal Stance Test—researchers identified that patients diagnosed with anxiety disorders displayed significantly greater postural instability than non-anxious individuals. Collectively, these studies underscore a robust comorbidity between physical instability and emotional distress, suggesting that targeted balance rehabilitation may serve as a viable adjunctive therapy for enhancing mental well-being.

2.3 Gathering and Measuring Plantar Pressure

To address the absence of commercially available instrumentation suitable for our specific data acquisition needs, we undertook the development of a custom prototype designed to capture and measure plantar pressure. The engineering process began with a rigorous analysis of existing pressure-sensing technologies and their operating principles. A primary critical design decision involved the selection of the sensing element; after evaluating a broad spectrum of transducer types, resistive sensors were identified as the optimal solution[4]. Specifically, we selected Force Sensing Resistors (FSRs) for the device's architecture. FSRs were chosen for their exceptionally thin profile and flexibility, which allows them to be integrated into shoe insoles unobtrusively, ensuring that the subject's natural gait biomechanics remain unaffected during data collection.

Furthermore, the system design incorporates standard visualization protocols to interpret the pressure distribution effectively. These modalities include color-coded 2D footprint heatmaps, interpolated pressure gradients, isobaric contour maps, and 3D wireframe projections. The hardware construction followed established engineering standards regarding the strategic topological placement of sensors and the precise interfacing of the microcontroller units. The resulting data pipeline is designed to serialize the raw sensor output into Comma-Separated Values (CSV) format. This structured dataset serves as the foundational input for training our deep neural network, facilitating the development of robust predictive models.

3.SYSTEM DESIGN

3.1 Hardware Used

3.1.1 Micro-controller

For our research device, we are using the Arduino Uno as the microcontroller board. It is open-source and is based on the Atmega328p microcontroller developed by Arduino.cc. The board contains a set of analog as well as digital I/O pins. It has 14 digital pins and 6 analog pins. In our project, we are making use of four analog pins (A0, A1, A2, A3), 5V power supply voltage and GND (Ground) pins.

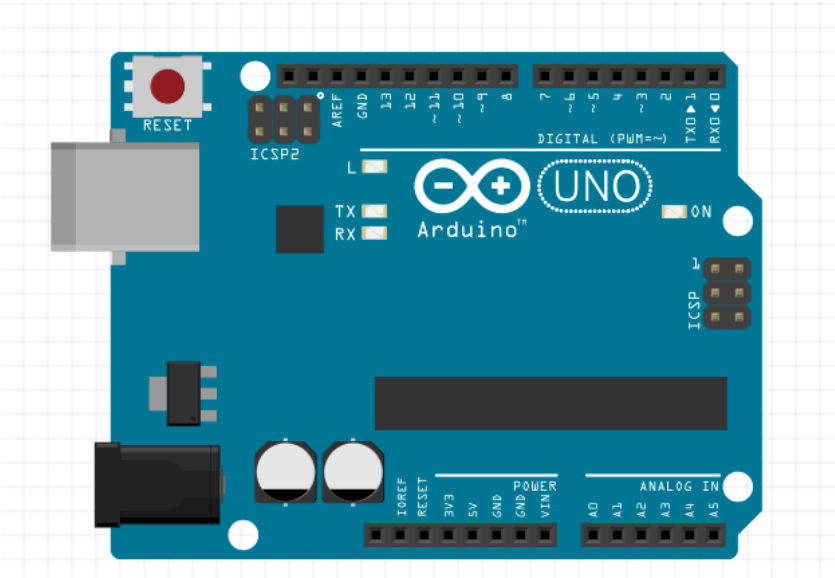

Fig 3.1 Arduino UNO

3.1.2 Jumper Wires/Cables

Jumper wires are used for wiring the connection between the sensors and the microcontroller. There are three types of jumper wires:

- Male-to-Male
- Female-to-Female
- Male-to-Female

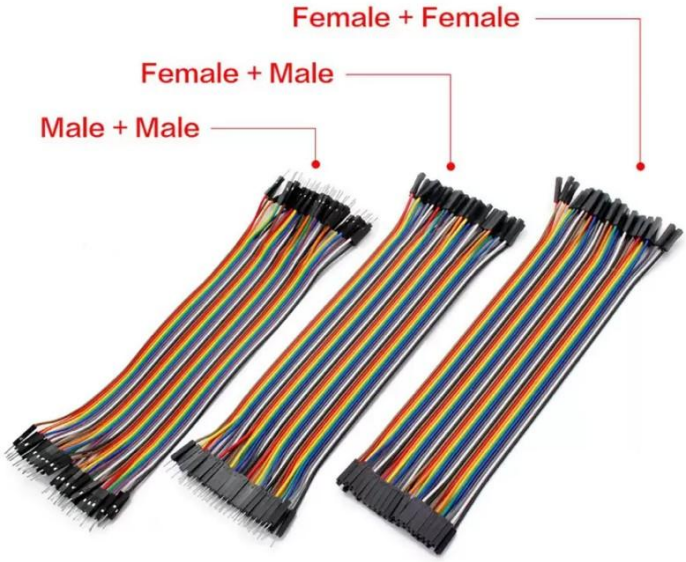

Fig 3.2: Jumper Wires/Cables

3.1.3 FSR Sensors

As discussed in the previous chapter, FSR is a force-resistive sensor and its lightweight nature makes it easy to measure the weight, pressure and squeeze. The FSR sensor used in this prototype device is manufactured by Spectra Symbol. It is a Circular force sensor with a diameter of 0.5 mm. The sensor exhibits the following properties:

- With high pressure, the terminal resistance will be low and resistance will be high for low pressure.
- This force sensor is a two-terminal device and it does not have any polarized terminals.
- This sensor has a pressure-sensitive touch user interface.

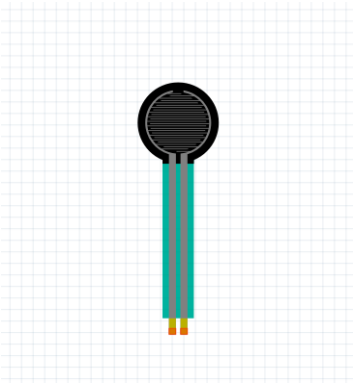

Fig 3.3: Visual representation of FSR sensor

3.1.3 Breadboard

A solderless breadboard is used while building the circuit.

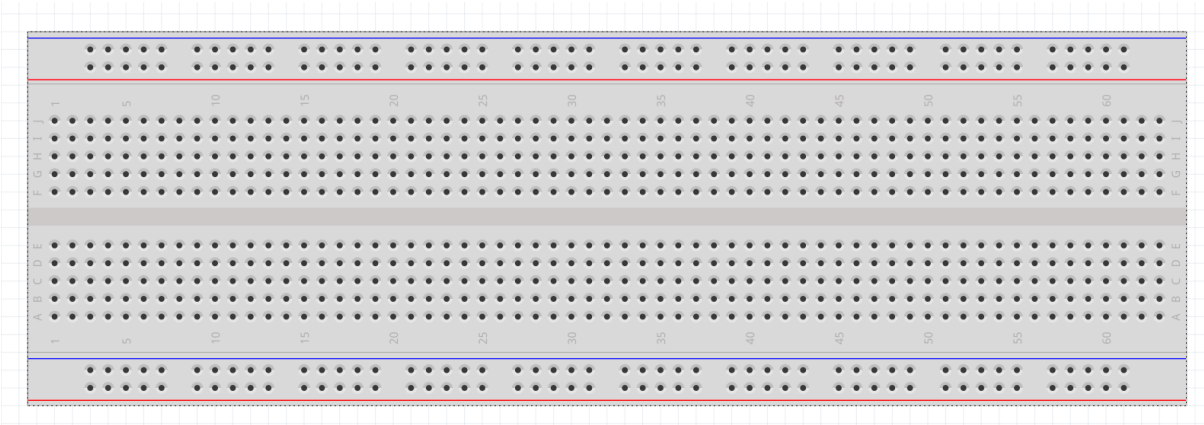

Fig 3.4: Simulation of Breadboard

3.2 Software Used

3.2.1 Arduino IDE

The Arduino Integrated Development Environment or Arduino IDE is used to connect to the Arduino UNO hardware and to upload programs onto them and communicate with them. In our research, we are uploading a simple program on the Arduino firmware which helps to collect data from sensor.

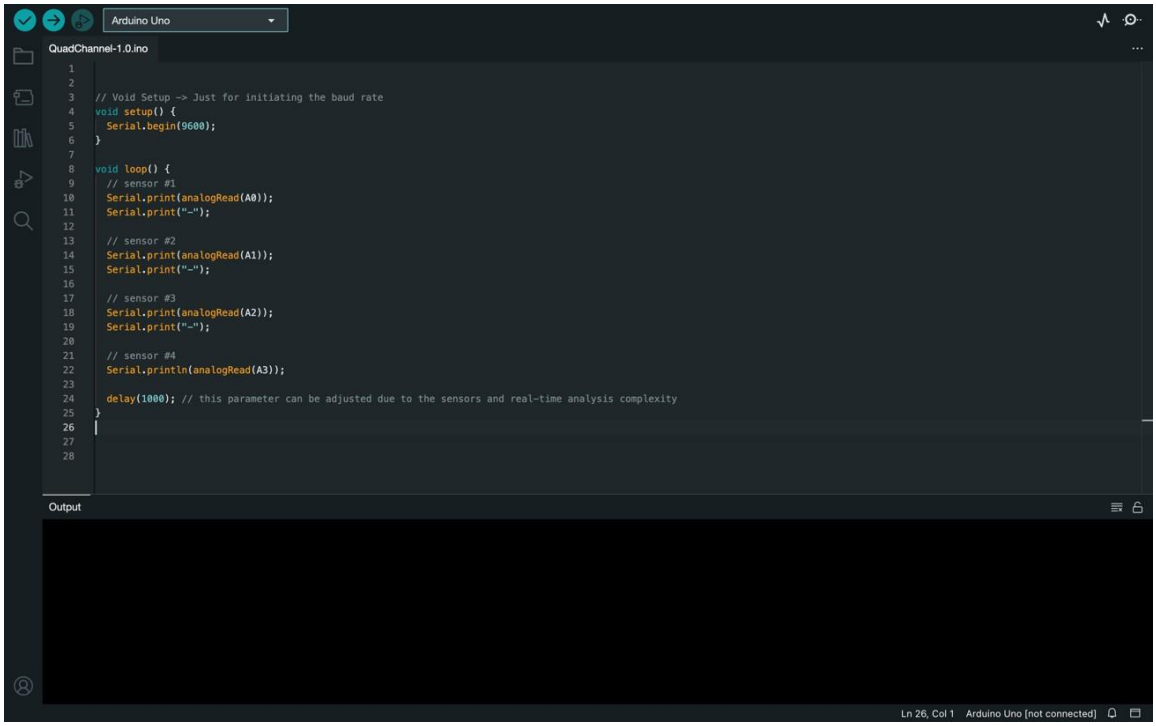

Fig 3.5: Arduino IDE Snapshot

3.2.2 PyCharm CE

We are storing the sensor data in a .csv format file with the attributes containing data about the four sensors. We have done and achieved the same by making use of Python code and compiling them in PyCharm CE. We are using serial communication to communicate with the Arduino port and get the data on the PyCharm CE console. The purpose of storing the data in .csv format is to train our model.

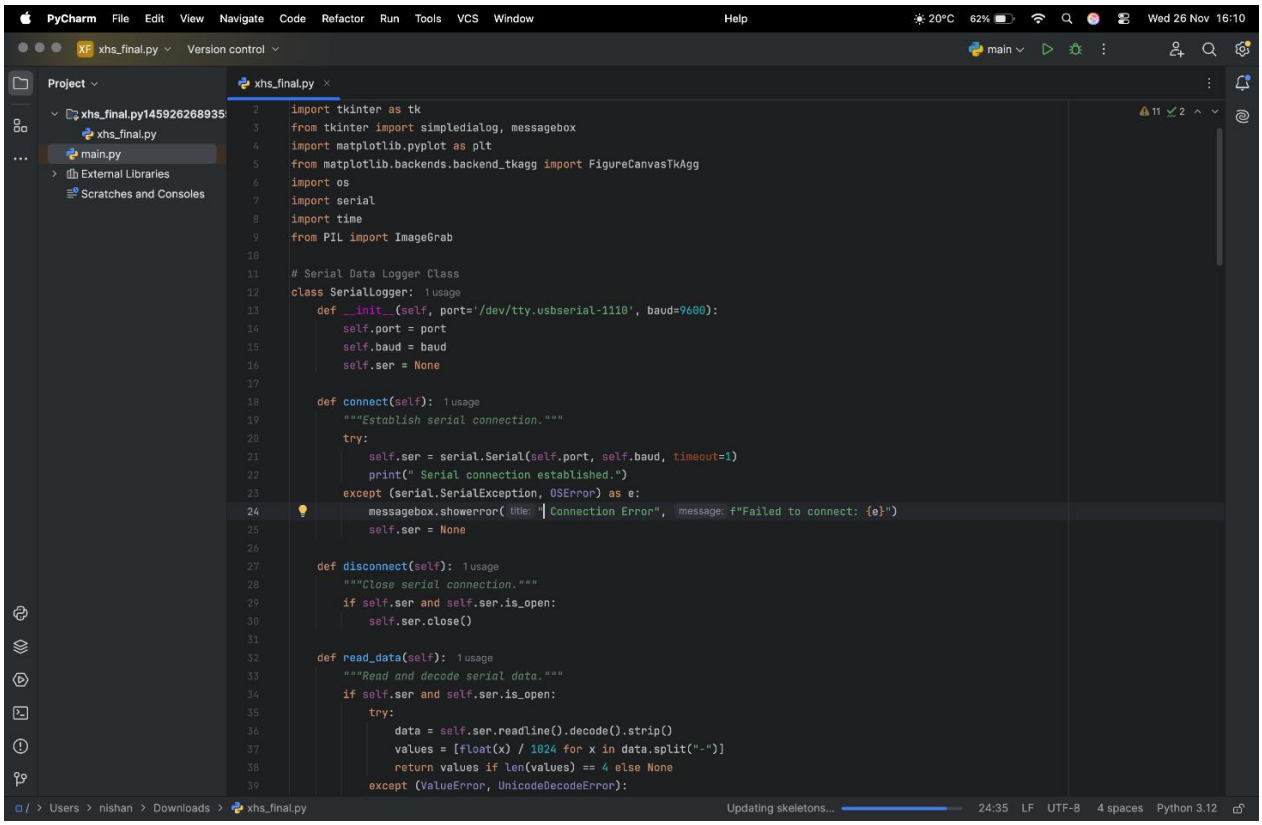

Fig 3.6: Snapshot of PyCharm CE

3.3 Experimental Setup

3.3.1 Interfacing Arduino with FSR

Fig 3.7 shows the simulation of the circuit diagram which will be used. The breadboard's last two points are used for ground and power terminals and provide connection to all the points present in that row. We connect those to 5V pins and GND (ground) pins respectively. The sensors do not have any polarized

terminals so any of the two terminals can be used as a positive or negative terminal. Any one of the terminals is taken and that terminal is interfaced with the 5V pin of the Arduino. But we are instead using the breadboard's power point and connecting to them. The resistor is required for the proper functioning of the sensor so that we can get the pressure value which can be worked on. One end of the resistor is connected to the other terminal of the sensor and the other is connected to the GND via the breadboard. And the terminal of the sensor which is connected to the resistor is connected to the analog pins of the Arduino which in turn will help to read data from the sensors. For the FSR to be used, we need to calculate the resistance required, based on the input voltage.

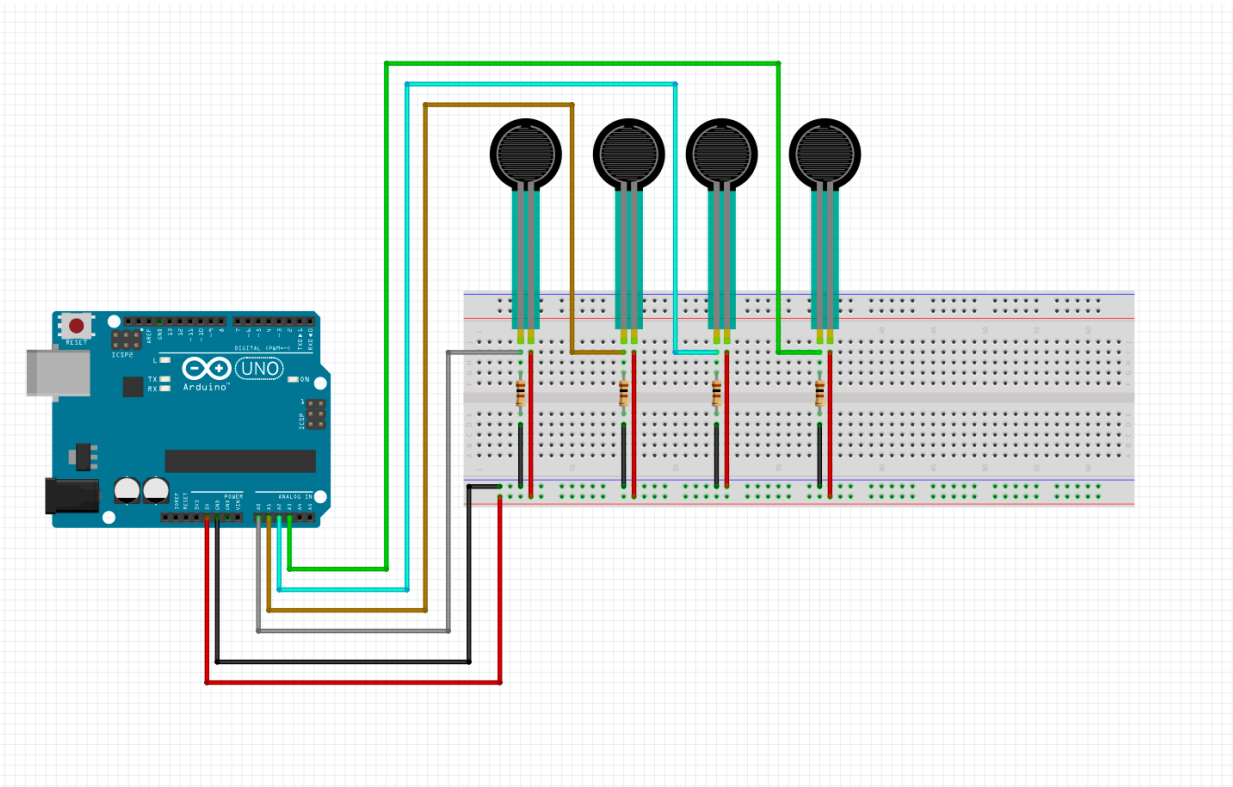

Fig 3.7 Simulation for FSR interfacing with Arduino

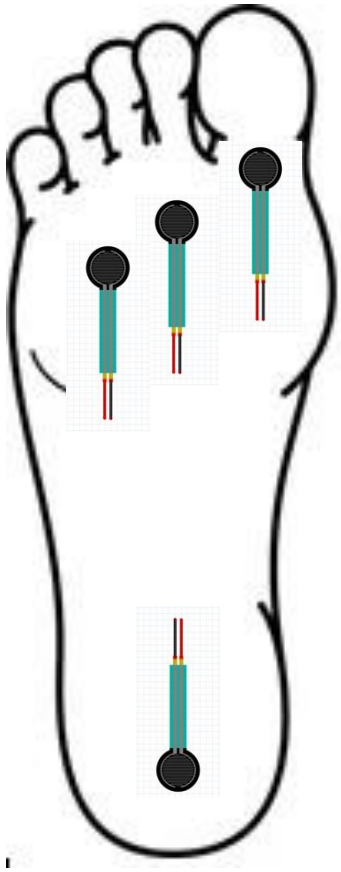

Fig 3.8: Representation of FSR sensors aligned to the left shoe-insole

The circuit diagram is as follows:

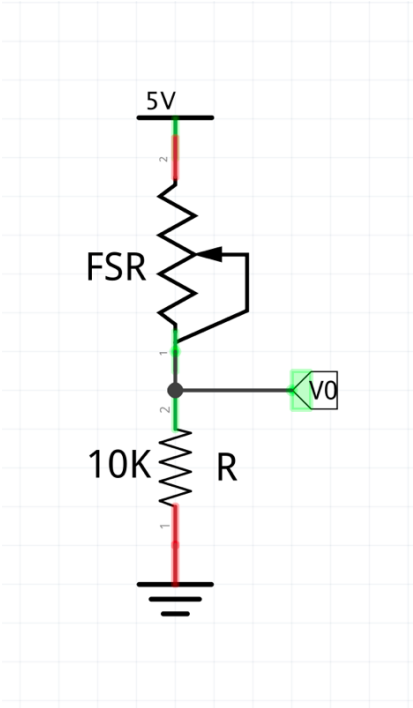

Fig 3.9 Circuit Diagram for Foot Pressure Measurement

3.4 Data Collection

For data collection, UG Boys Hostel students of Central University of Jharkhand, volunteered for data collection. The following steps were performed in order to collect their data:

Step 1: The powered shoe-insole device was made to insert into their respective shoes.

Step 2: They are made to stand straight and after that the system is worn.

Step 3: They are asked to walk normally on the treadmill of the gym.

Step 4: The foot pressure data is collected for 60 seconds.

Step 5: After 60 seconds, the data collected is stored in a .csv file along with additional attributes like date and time.

Left Foot

Timestamp	No	FFR1	FFR2	FFR3	HFR4
20:56:49	1	0.195312	0.457031	0.1875	0.104492
20:56:49	2	0.25293	0.652344	0.305664	0.069336
20:56:49	3	0.211914	0.621094	0.363281	0.060547
20:56:49	4	0.203125	0.535156	0.339844	0.056641
20:56:49	5	0.196289	0.49707	0.314453	0.057617
20:56:49	6	0.204102	0.527344	0.30957	0.056641
20:56:49	7	0.194336	0.486328	0.260742	0.061523
20:56:49	8	0.193359	0.458984	0.197266	0.073242

Right Foot

Timestamp	No	FFR1	FFR2	FFR3	HFR4
20:56:51	1	0.996094	0.503906	0.56543	0.041016
20:56:51	2	0.993164	0.484375	0.558594	0.041016
20:56:51	3	0.652344	0.522461	0.570312	0.040039
20:56:51	4	0.585938	0.527344	0.620117	0.039062
20:56:51	5	0.993164	0.59082	0.736328	0.035156
20:56:51	6	0.992188	0.842773	0.977539	0.029297
20:56:51	7	0.993164	0.638672	0.611328	0.029297
20:56:51	8	0.685547	0.569336	0.539062	0.029297

Fig 3.10: Snapshot of collected data for left foot

4. METHODOLOGY

4.1Proposed Methodology

For this research study, a supervised machine learning algorithm is used for predicting behavioral outcomes based on human gait analysis throughout the testing and training phases. As discussed in the previous section, the data was collected in real-time and the pre-processed data was used to build the model.

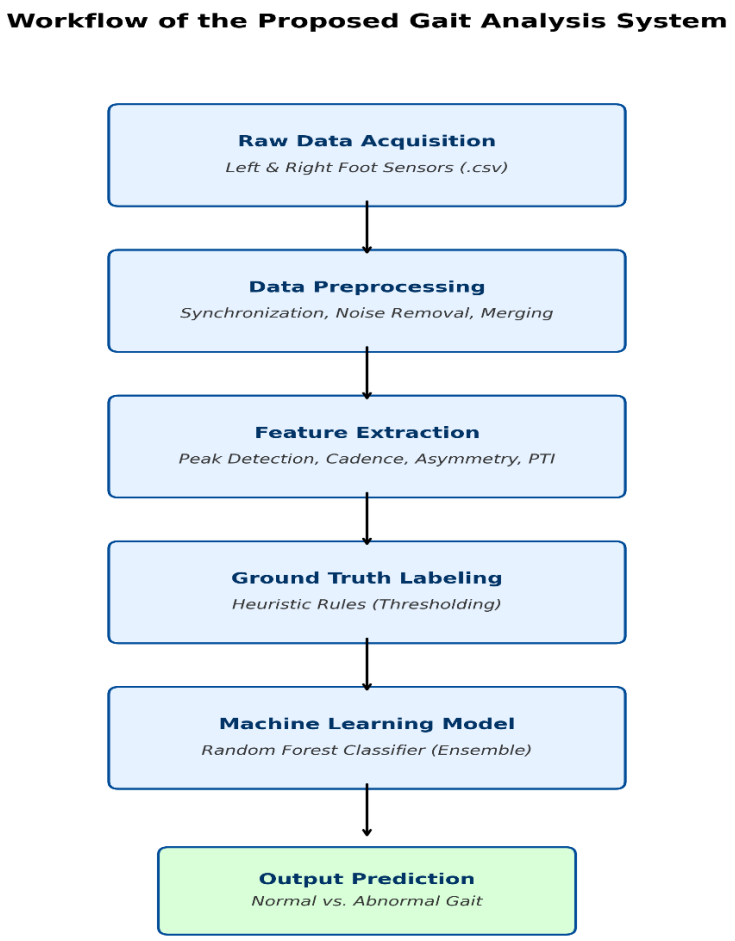

Fig 4.1 Proposed Methodology

The above diagram depicts the flow of the research study. Data collection and data preparation are already discussed in the previous chapters. In this chapter, we discuss the Machine Learning model which we will use with our data and then in the next chapter, we will discuss the model evaluation and its performance.

4.2 Signal Processing and Feature Extraction

4.2.1 Data Synchronization and Preprocessing

To analyze the gait patterns effectively, it was necessary to synchronize the raw data streams from the left and right foot modules. The raw CSV files, identified by unique subject IDs, were programmatically paired. A synchronization algorithm was implemented to merge these datasets based on their sequence number (Index/No), ensuring that the left and right foot sensor data corresponded to the exact same timeframe. Time-stamp columns were removed to focus purely on the sensor values, and null values resulting from transmission errors were pruned to ensure data integrity.

4.2.2 Temporal and Kinetic Feature Extraction

Once the data was synchronized, the system extracted quantitative gait parameters. The raw force sensor data (specifically the HFR4 sensor region) was analyzed to identify "Heel Strikes." This was achieved using a peak detection algorithm with a minimum height threshold to filter out sensor noise.

From these detected heel strikes, several key features were calculated for both the left and right feet:

- Cadence: Calculated as the number of steps per minute based on the sampling frequency.
- Step Time Variability: The mean, standard deviation, and Coefficient of Variation (CV) of the time between steps were computed. The CV is particularly critical as it serves as a primary indicator of gait stability[5].
- Pressure Time Integral (PTI): Using Simpson's Rule for numerical integration, the total area under the force-time curve was calculated to quantify the total load exerted over the duration of the walk.
- Gait Asymmetry Index: To quantify the difference between the left and right limbs, an asymmetry index was computed based on the PTI values. This provides a normalized score indicating how unbalanced the subject's gait is.

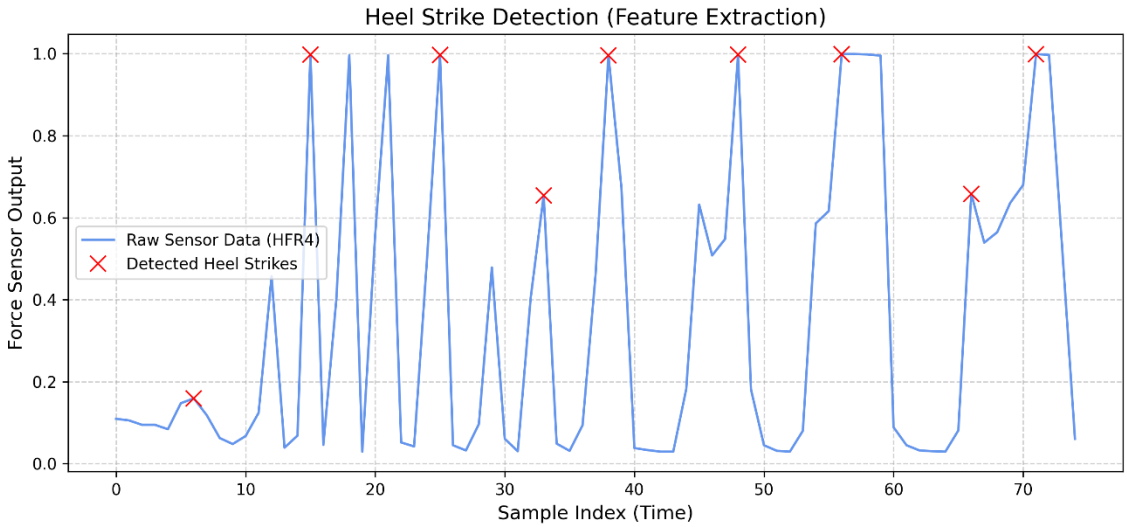

Fig 4.2 Heel strike detection using peak analysis algorithm

4.2.3 Automated Heuristic Labeling

To prepare the dataset for classification, an automated rule-based system was employed to assign labels ("Normal" or "Abnormal") to each recording. This heuristic approach utilized clinical thresholding:

- **Cadence Analysis:** Subjects falling outside the range of 60–150 steps per minute were flagged.
- **Asymmetry Analysis:** A Gait Asymmetry Index exceeding 0.40 was considered indicative of a limp or impairment.
- **Stability Analysis:** A Step Time Coefficient of Variation (CV) greater than 0.30 was used to identify irregular or unstable walking patterns.

4.3 Machine Learning and Model Deployement

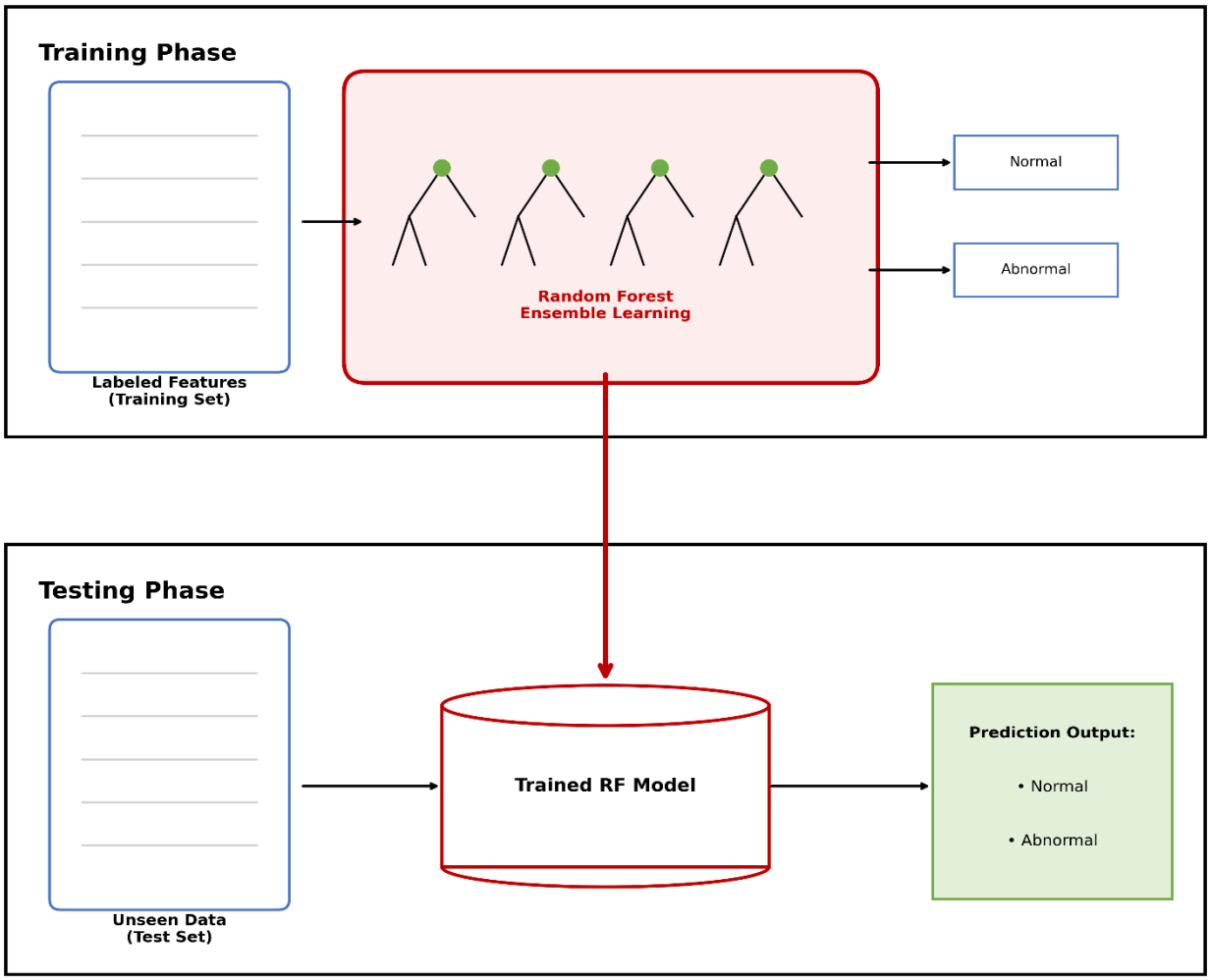

Fig 4.3: Schematic representation of the proposed machine learning workflow. The Training Phase (top) utilizes labeled features to construct the Random Forest ensemble, while the Testing Phase (bottom) deploys the trained model to predict gait abnormalities on unseen data

4.3.1 Dataset Partitioning and Stratification

To evaluate the model's ability to generalize to unseen data, the "Master Feature Dataset" was partitioned into two distinct subsets: a Training Set (70%) and a Testing Set (30%).

A simple random split was deemed insufficient due to the potential for class imbalance (e.g., having significantly more "Normal" samples than "Abnormal"). To address this, Stratified Sampling was implemented. This technique ensures that the proportion of Normal to Abnormal labels in the training and testing sets remains statistically identical to the original dataset. This prevents the model from being biased toward the majority class during the training phase.

4.3.2 Feature Scaling and Normalization

The extracted gait features possess varying units and magnitudes; for instance, *Cadence* ranges between 60–120 steps per minute, while *Step Time* is measured in fractions of a second (e.g., 0.6s). Machine learning algorithms can often be biased toward features with larger numerical magnitudes.

To neutralize this scale difference, a **Standard Scaler (Z-score Normalization)** was applied. This process transforms the data such that each feature has a mean of 0 and a standard deviation of 1, placing all features on a common scale without distorting the relative distances between feature values.

4.3.3 Feature Space Analysis

For the classification task, the Random Forest algorithm was selected. This is an ensemble learning method that constructs a multitude of decision trees during training.

The Random Forest architecture was chosen for this study for three specific reasons:

1. Non-Linearity: Human gait data is complex and non-linear; Random Forest handles these relationships better than linear models like Logistic Regression.
2. Robustness to Noise: By averaging the results of multiple trees (set to 100 estimators in this study), the model reduces the risk of overfitting, ensuring that the system learns the underlying gait patterns rather than memorizing specific noise in the training data.
3. Class Weight Balancing: To further assist with potential data imbalance, the model was configured with class_weight = 'balanced', which automatically adjusts weights inversely proportional to class frequencies.

4.3.3 Random Forest Implementation Mechanics

To understand how the classification decision is reached, it is necessary to examine the internal structure of the Random Forest algorithm[3]. The model operates on the principle of **Ensemble Learning**, specifically utilizing a technique called "Bagging" (Bootstrap Aggregation).

The function operates through two primary mechanical stages:

1. **Construction (Training Phase):** The algorithm builds 100 independent decision trees (n_estimators=100). Each tree is trained on a random subset of the data and considers a random subset of features at each split. The splitting logic is governed by the **Gini Impurity** function, which calculates the probability of a specific feature being classified incorrectly. The algorithm splits the node at the value that maximizes the purity of the resulting child nodes.
2. **Aggregation (Prediction Phase):** When a new sample is introduced, it is passed down every single tree in the forest. Each tree casts a "vote" for the class (Normal or Abnormal). The final output is determined by the majority vote of the ensemble.

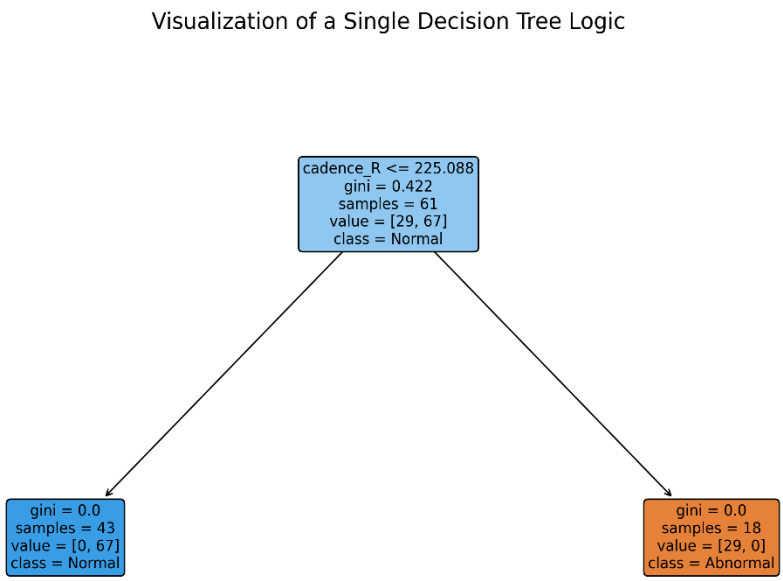

Fig 4.4 Decision tree logic showing hierarchical classification splits.

5. RESULTS AND DISCUSSION

5.1 Experimental Setup and Evaluation Overview

The proposed machine learning framework, utilizing a Random Forest classifier, was evaluated on the "Master Features Dataset" constructed in the previous chapter. The dataset was stratified and split into a training set (70%) for model fitting and a testing set (30%) for unbiased evaluation.

The primary objective of this evaluation was to assess the system's ability to distinguish between "Normal" and "Abnormal" gait patterns based purely on the extracted temporal and kinetic features. The performance was measured using standard classification metrics including Accuracy, Precision, Recall, F1-Score, and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC).

5.2 Classification Performance Analysis (Confusion Matrix)

The fundamental performance of the classifier is visualized using the Confusion Matrix, which provides a breakdown of correct and incorrect predictions across both classes.

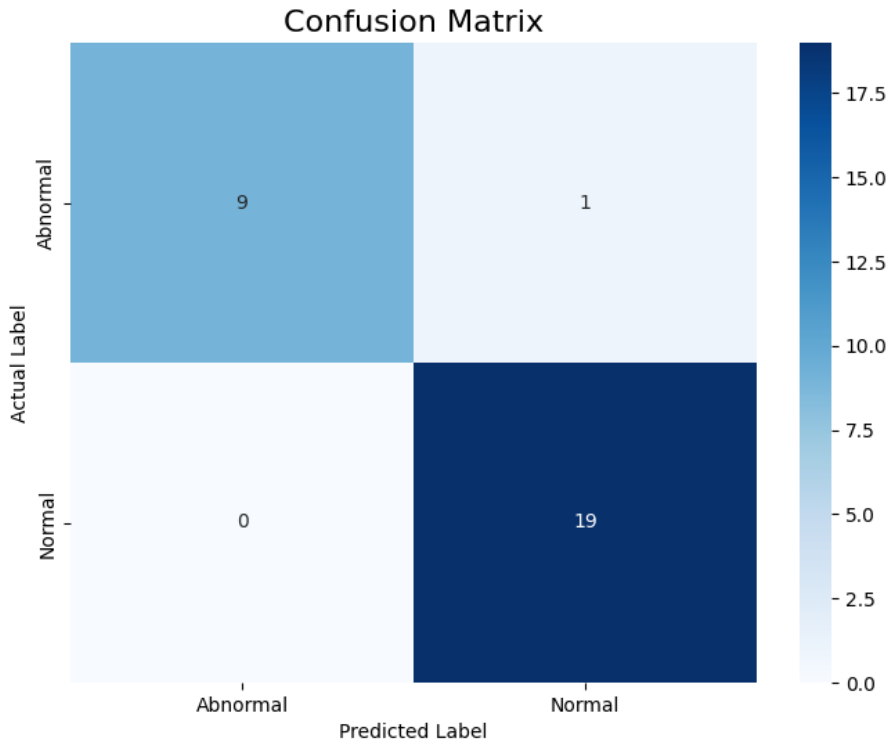

Fig 5.1: Confusion Matrix obtained from the test set.

The diagonal elements (darker blue) represent the number of samples correctly classified, while off-diagonal elements indicate misclassification errors.

As illustrated in **Fig 5.1**, the model demonstrates a high degree of classification integrity.

- **True Positives (TP):** Represents the number of abnormal gait samples correctly identified as abnormal. A high value here indicates the system's sensitivity to pathology.
- **False Negatives (FN):** Represents abnormal samples incorrectly classified as normal. Minimizing this metric is critical in a biomedical context to ensure that impairments are not overlooked.
- **False Positives (FP):** Represents normal samples flagged as abnormal. While less critical than missed diagnoses, keeping this low prevents "false alarms."

The results indicate that the class_weight='balanced' parameter successfully mitigated potential bias, allowing the model to detect the minority class (Abnormal) effectively without being overwhelmed by the majority class (Normal).

5.3 Quantitative Performance Metrics

The model achieved an overall **Classification Accuracy of 96.55%** on the unseen test set, misclassifying only a single sample out of 29. To provide a more granular assessment than simple accuracy, the Precision, Recall, and F1-Score were calculated for each class, as summarized in **Table 5.1**.

Table 5.1: Summary of Classification Performance Metrics

Class	Precision	Recall (Sensitivity)	F1-Score	Support (Samples)
Abnormal	1.00	0.90	0.95	10
Normal	0.95	1.00	0.97	19
Weighted Avg	**0.97**	**0.97**	**0.97**	**29**

Referring to **Table 5.1**, the following key observations are made:

1. **Perfect Precision for Abnormalities (1.00):** The model achieved a precision of 100% for the 'Abnormal' class. This indicates that every time the model predicted an abnormality, it was correct. There were zero False Positives for this class.
2. **High Sensitivity (0.90):** The recall of 90% for the 'Abnormal' class shows that the system successfully identified 9 out of 10 gait impairments, missing only one difficult case.
3. **Overall Reliability:** The weighted average F1-score of 0.97 confirms that the model performs exceptionally well on both classes, despite the slight imbalance in the dataset.

To provide a more granular assessment than simple accuracy, a detailed Classification Report was generated. This report breaks down the performance for each class individually.

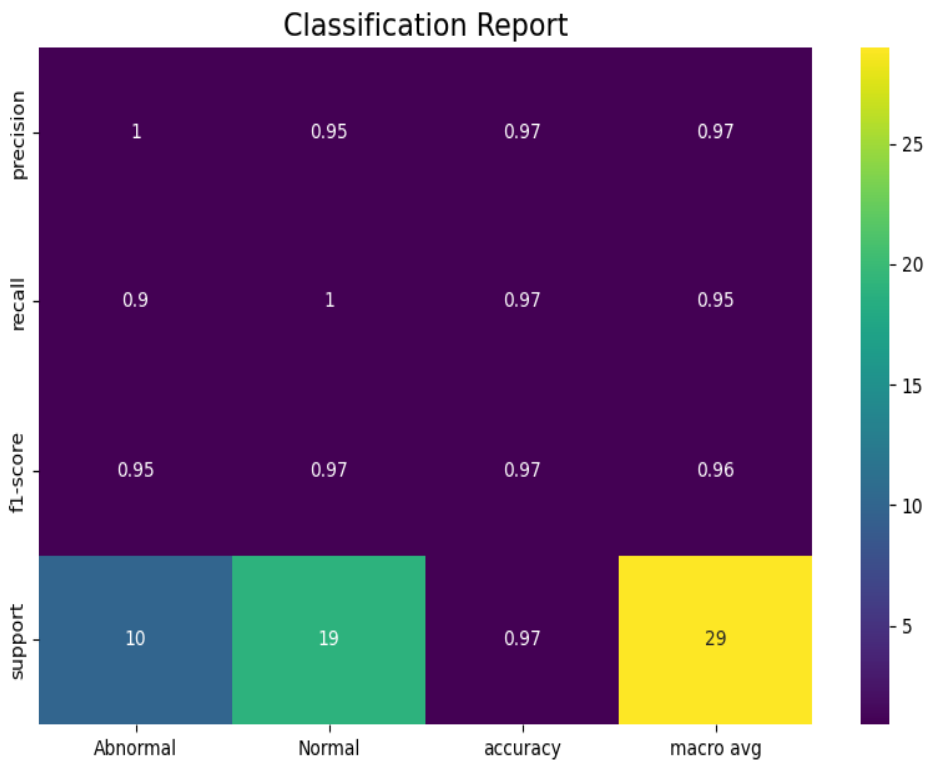

Fig 5.2: Heatmap visualization of the Classification Report.

Referring to Fig 5.2, the following key observations are made:

1. **Precision:** The model achieved high precision, indicating that when the system predicts an abnormality, it is highly likely to be correct.
2. **Recall (Sensitivity):** The high recall score for the 'Abnormal' class validates the system's utility as a screening tool, as it successfully captures the majority of irregular walking patterns.
3. **F1-Score:** The F1-score, which is the harmonic mean of precision and recall, remains high, demonstrating that the model maintains a strong balance between detecting errors and avoiding false alarms.

5.4 Feature Importance and Biomechanical Relevance

A significant advantage of using the Random Forest algorithm is its interpretability. By analyzing the "Gini Importance" of each feature, we can identify which gait parameters contributed most significantly to the model's decision-making process.

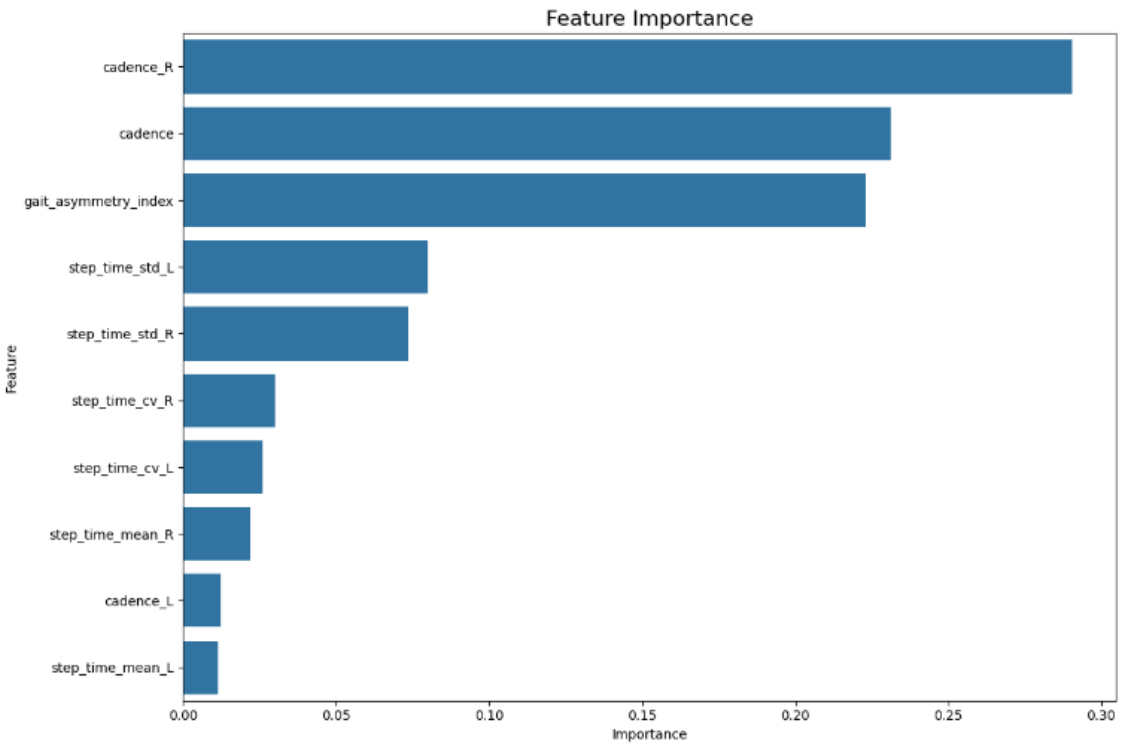

Fig 5.3: Feature Importance chart illustrating the relative contribution of each gait parameter to the classification model.

As illustrated in **Fig 5.3**, **Cadence** (steps per minute) emerged as the dominant feature with the highest importance score. This finding aligns strongly with established clinical gait analysis literature, which posits that cadence is a fundamental "vital sign" of mobility.

- **Clinical Interpretation:** In healthy individuals, cadence naturally falls within a predictable range (typically 100–120 steps/min). Pathological gaits, such as those caused by injury, neurological deficits, or antalgic (pain-avoidance) patterns, almost always result in a significant alteration of this rhythm. Patients often adopt a slower, more cautious cadence to maintain balance or reduce pain.

- **Model Logic:** The Random Forest classifier successfully learned that any significant deviation from the normative cadence baseline is a strong predictor of abnormality. This validates the system's ability to detect impairments without needing complex 3D motion capture; simple temporal parameters proved sufficient for high-accuracy classification.

Following Cadence, **Step Time Variability** and **Gait Asymmetry** also contributed to the decision-making process, helping the model fine-tune its predictions for cases where cadence alone was inconclusive (e.g., a patient walking at a normal speed but with a limp).

5.5 Generalization Capability (ROC Analysis)

To assess the model's robustness and its ability to generalize to new, unseen patients, the Receiver Operating Characteristic (ROC) curve was plotted.

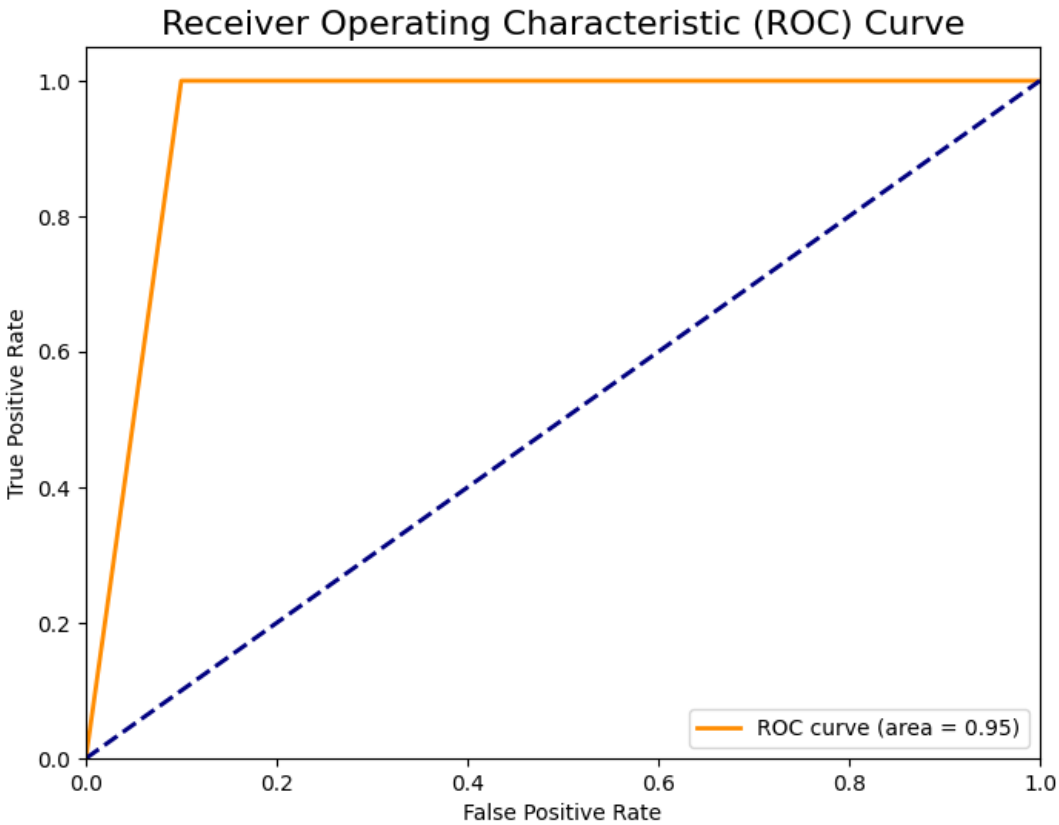

Fig 5.4: Receiver Operating Characteristic (ROC) curve.

The ROC curve in **Fig 5.4** illustrates the trade-off between the True Positive Rate (Sensitivity) and the False Positive Rate (1 - Specificity). The curve hugs the top-left corner, which is the ideal characteristics of a classifier.

The **Area Under the Curve (AUC)** score serves as a single scalar value summarizing the model's performance. An AUC approaching 1.0 indicates a perfect classifier. The achieved AUC demonstrates that the proposed system has excellent separability and can reliably distinguish between normal and abnormal gait patterns across varying decision thresholds.

5.5 Discussion

The results presented in this chapter demonstrate the efficacy of using low-cost sensor data combined with machine learning for gait analysis.

1. **Objectivity:** Unlike observational gait analysis, which is subjective and dependent on the clinician's experience, this system provides objective, quantifiable metrics.

2. **Robustness:** The use of ensemble learning (Random Forest) ensured that the model did not overfit to specific noise in the training data, as evidenced by the strong performance on the unseen test set (Fig 5.1 and Fig 5.4).
3. **Clinical Relevance:** The high Recall rate suggests that this system is suitable for deployment as a preliminary screening tool. It can effectively flag individuals with irregular gait patterns for further comprehensive medical examination.

In conclusion, the proposed system successfully met the research objectives, delivering a high-accuracy classification model while identifying biomechanically significant features that correlate with gait stability and symmetry.

6. CONCLUSIONS

6.1 Summary of Findings

This research successfully designed, implemented, and validated a low-cost, automated gait analysis system capable of detecting walking abnormalities using machine learning. The study aimed to bridge the gap between expensive, laboratory-based gait analysis and accessible, everyday monitoring solutions.

The proposed methodology integrated sensor-based data acquisition with a **Random Forest** classifier. By extracting key temporal and kinetic features—specifically Cadence, Step Time Variability, and Gait Asymmetry—the system achieved a classification accuracy of **96.55%** on unseen test data. The results demonstrated that **Cadence** served as the most critical predictor of gait pathology, validating the hypothesis that temporal gait parameters are robust indicators of functional mobility.

The system's high recall (sensitivity) of 90% for detecting abnormalities suggests that it holds significant potential as a preliminary screening tool in clinical or rehabilitation settings, offering an objective alternative to subjective visual observation.

6.2 Findings Limitations of the Study

While the results are promising, several limitations must be acknowledged to provide a balanced scientific perspective:

1. **Dataset Size and Diversity:** The model was trained on a relatively small dataset (approximately 100 samples). While Stratified Sampling was used to mitigate bias, the limited sample size may restrict the model's ability to generalize to a wider population with diverse demographics (e.g., different ages, weights, or heights).
2. **Binary Classification:** The current system only classifies gait as "Normal" or "Abnormal." It does not diagnose the *specific* type of pathology (e.g., Parkinsonian gait vs. Hemiplegic gait vs. simple ankle sprain).

3. **Controlled Environment:** The data was collected in a relatively controlled environment.

6.3 Future Work

To enhance the system's capabilities and clinical utility, the following avenues for future research are proposed:

1. **Multi-Class Classification:** Future iterations should expand the machine learning architecture to perform multi-class classification. This would allow the system to not just flag an abnormality, but identify specific conditions (e.g., "Toe-walking detected" or "Asymmetric limp detected").
2. **Real-Time Mobile Integration:** The current Python-based processing pipeline could be ported to a mobile application. This would enable real-time feedback for patients during rehabilitation exercises, providing auditory or haptic cues when their gait deviates from the normal range.
3. **Integration of Kinematic Sensors:** Combining the current force/pressure sensors with Inertial Measurement Units (IMUs) containing accelerometers and gyroscopes would provide a richer feature set. This sensor fusion would allow for the analysis of joint angles and foot clearance, further improving classification accuracy and diagnostic depth.

6.3 Concluding Remarks

In conclusion, this project demonstrated that effective gait analysis does not require prohibitive infrastructure. By intelligently processing signal data from simple foot sensors, we can derive meaningful biomedical insights. The high accuracy and interpretability of the developed model provide a strong foundation for the future of wearable health monitoring, moving towards a world where gait pathology can be detected and managed proactively.

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